MARVEL TECH
LAJOYA City Budget App
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Part 2: Planning and Design Lajoya City Budget App

1. Introduction

App Name: Lajoya City
Icon Design: The icon features a vibrant, circular gold coin base representing financial prosperity. Embossed on the coin is the minimalist silhouette of a skyscraper, symbolizing the building of one's wealth. The color palette utilizes Emerald Green (#2ECC71) for growth and Goldenrod (#F1C40F) for value.

Overview
Lajoya City is a mobile application designed to alleviate the psychological stress of financial tracking. By merging the strict budgetary logic of a traditional expense manager with the rewarding feedback loops of a city building game, the app transforms mundane data entry into a constructive experience. As established in the research phase, users are more likely to maintain a habit when they see a visual representation of their progress (Chou, 2019). *Lajoya City* provides this through a dynamic Progress Dashboard and a unique Streak Multiplier gamification engine.

Innovative Features:

- The Streak Multiplier: Users who log expenses for five consecutive days earn Multiplier Points. These points unlock high-tier visual assets, such as Premium Icons for categories (e.g., a Golden Shopping Bag instead of a standard one).
- Contextual Visual Nudges: Unlike standard apps that only show numbers, *Lajoya City* uses an adaptive UI. If a user is under budget, the background of the dashboard reflects a Sunny theme; if they exceed 90% of their budget, the UI shifts to a Sunset amber warning.

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2. Detailed Requirements

2.1 User Authentication and Security

The app will implement a secure local login system.

- Registration: Users must provide a unique username and a password. The password must be at least 8 characters long and include a number to ensure data security.
- Login: The app will validate credentials against a local SQLite/Room database. If a user enters incorrect details three times, a Cool down timer will trigger to prevent brute-force attempts.

2.2 Category Management

- Customization: Users must be able to create, edit, and delete expense categories. Each category will be associated with a specific icon and a budget limit.
- Validation: The app will prevent the creation of duplicate category names to maintain data integrity.

2.3 Expense Entry and Multimedia

- Data Fields: Every entry requires an Amount (Decimal), Date (Date Picker), Description (String), and Category (Spinner selection).
- Camera Integration: Utilizing the Android Camera X API, users can capture a photo of their receipt. The app will store the image URI in the local database and display a thumbnail preview within the entry form.

2.4 Budgeting and Limit Logic

- Monthly Totals: The user can set a Master Budget for the month.
- Category Limits: Specific limits can be set for individual categories. If the Entertainment limit is R5000, the app must calculate the remaining balance every time a new entry is added.

2.5 Data Persistence

All user data, including category definitions, expense entries, and user profiles, will be saved to a Room Persistence Library (an abstraction over SQLite). This ensures that data is retained even if the app is closed or the device is restarted.

2.6 Analytics and Visualizations

- Daily Spending Graph: Using the MPAndroidChart library, the app will render a line graph showing daily spending. Users can filter this graph to show the last 7, 14, or 30 days.

- Progress Dashboard: A central hub showing a Master Progress Bar and smaller Category Progress Bars. These bars will be color coded (Green for <50%, Amber for 50-80%, Red for >80% spent).

2.7 Gamification and Rewards

- Badge System: The app will check for Achievement Triggers (e.g., First Expense Logged, Budget Hero). When a trigger is met, a Custom Toast or Dialog will appear awarding a badge.
- XP Logic: Users gain 10 Experience Points (XP) for every entry. Reaching a new level unlocks new UI themes.

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3. User Interface Design

3.1 Screen Mock ups and Descriptions

- Splash Screen: A branded screen showing the *Lajoya* icon and a loading bar. Purpose: To initialize database connections and GitHub Action environment checks.
- Auth Screen: A clean, centered layout with two EditText fields for Username and Password. Buttons for Login and Sign Up are clearly distinguished by color.
- Dashboard (Main Screen): This is the heart of the app.
  - *Top:* Total Monthly Budget vs. Spent (Large Circular Progress Bar).
  - *Middle:* The Daily Spending Graph.
  - *Bottom:* A horizontal scroll of Category Progress Bars.

- Expense Screen: A form-based layout. It includes a Floating Action Button (FAB) that opens the camera to Attach Receipt.
- History List Screen: A RecyclerView displaying all expenses. Each item shows the Category Icon, Date, and Amount. Tapping an item opens a Detail View where the user can see the full description and the receipt photo.

3.2 Navigation Diagram

The app will use a Bottom Navigation Bar for high-level navigation between the following destinations:

- Home: Dashboard and Analytics.
- History: List of all transactions and period selection.
- Categories: Management of spending limits and icons.
- Profile: Settings, Gamification Badges, and Logout.

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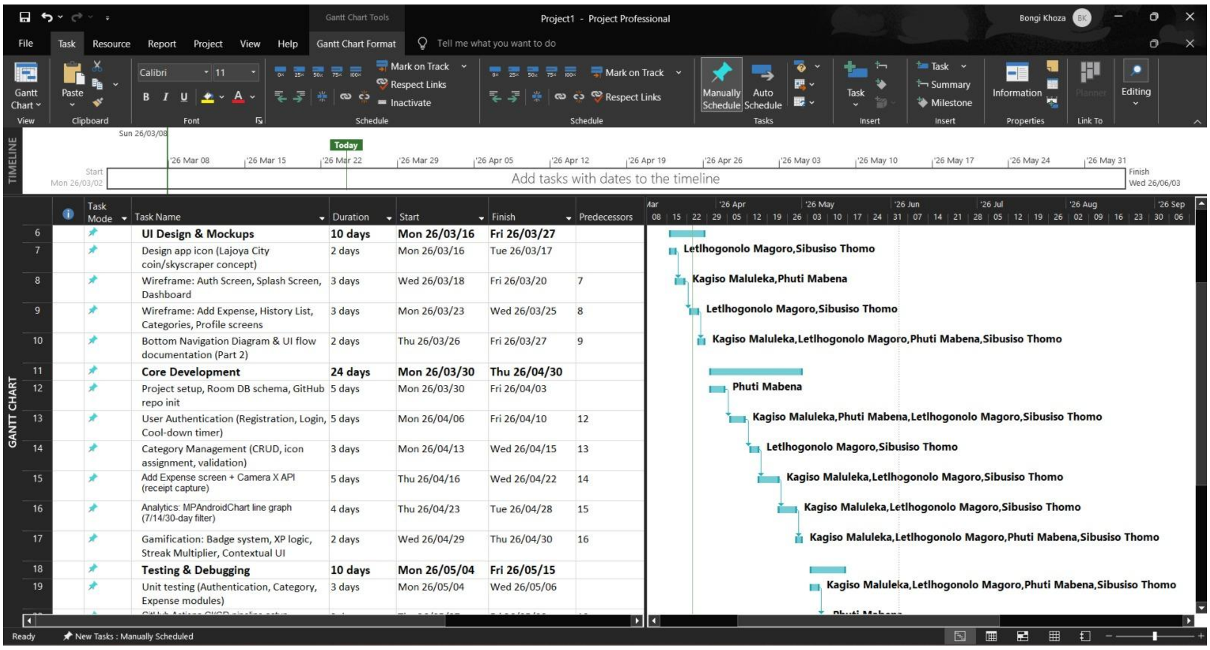
4. Project Plan

Milestone	Task Description	Duration
M1: Foundation	Project setup, Room Database schemas, and UI Wireframing.	Week 1
M2: Core Logic	Implementation of Authentication and Category Management.	Week 2
M3: Transactions	Add Expense functionality and Camera API integration.	Week 3
M4: Analytics	Implementation of MPAndroidChart for daily spending trends.	Week 4
M5: Gamification	Badge logic, XP calculation, and Progress Dashboard UI.	Week 5
M6: Quality	Unit Testing, GitHub Actions setup, and Final Bug Fixing.	Week 6



5. Conclusion

The design of *Lajoya City* is strategically crafted to solve the engagement problem in financial apps. By following the Research Design Build cycle, this project ensures that every feature from the receipt capture to the line graph serves a specific user need. The transition from the research of *Fortune City* and *Wallet* into the design of *Lajoya City* guarantees a product that is both functionally robust and psychologically rewarding.

6. References

- BudgetBakers. (2024). *Wallet: Why we focus on Financial Health*. [online] Available at: https://budgetbakers.com/features/ [Accessed 20 Mar 2024].
- Chou, Y-k. (2019). *Actionable Gamification: Beyond Points, Badges, and Leaderboards*. 2nd edn. Octalysis Media.
- Google Play. (2024). *Money Manager Expense & Budget*. [online] Available at: https://play.google.com/store/apps/details?id=com.realbyte.moneymanager [Accessed 20 Mar 2024].
- Smith, J. (2022). 'The Importance of White Space in Fintech Design', *Journal of Mobile UX*, 12(4), pp. 45-58.
